DNA *in Virtuo*
Visualization and Exploration of 3D Genomic Structures

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ABSTRACT
In this paper, we address the potential offered by Virtual Reality and scientific simulation for 3D modeling and immersive visualization of huge genomic sequences. Advanced work on 3D data modeling and structuring is proposed. In Bioinformatics, DNA sequences are often represented within linear format. However, they also have a three-dimensional structure potentially suitable for genomic analysis. The representation of such 3D structure allows biologists to observe and analyze genomes in an interactive way at different levels: from gene to chromosome. We developed a powerful software platform that provides a new point of view for sequences analysis: *ADN-Viewer*. Nevertheless, a classical eukaryotic chromosome of 40 million base pairs requires about 6 Gbytes of 3D data. In order to manage these huge masses of data in real-time, we designed various scene management algorithms and immersive human-computer interaction for *user-friendly* data exploration.

Keywords
Virtual Reality, Scientific Visualization, 3D Modeling, Human-Computer Interaction, Bioinformatics.

1. INTRODUCTION
Most biologists work on textual DNA files (i.e. sequence of the four known nucleotides A, C, G and T) that are limited to linear representation. Besides, such linearity offers only local and partial vision of these molecules (each letter in the sequence represents a nucleotide). However, within a cell, the DNA is supercoiled as a double-helix, resulting into complex trajectory defined in space. The 3D structure of such complex conformation is very important in many essential biological mechanisms, like replication, transcription or regulating. Thanks to 3D conformation models, we are able to construct the 3D trajectory of a DNA molecule given its textual sequence. Some softwares as DNATools[1] represent the 3D DNA trajectory but they are limited up to several hundreds of nucleotides. Others (RasMol[2], Swiss-PDB Viewer [4]…) display very well data contained in PDB[3] files and do not construct any trajectory. In fact, a DNA sequence contains a huge amount of nucleotides: bacteria's or archaea's chromosomes contain between 1 and 10 million nucleotides, while eukarya's (animals, plants, humans...) chromosomes usually contain several million up to several hundred million nucleotides. In this context, the visualization of a whole chromosome requires a lot of graphic and memory resources. Our main aim was to design a powerful and *user-friendly* visualization software tool that renders whole molecules in real-time, taking into account such data mass.

The 3D information, resulting from modeling construction, could either be visualized or post-processed. The 3D visualization, contrary to the textual one, offers a global vision of the molecules and is achieved through a software called *ADN-Viewer* [2] developed at LIMSI-CNRS. Presently, *ADN-Viewer* can load and visualize multiple sequences of about several ten million nucleotides (depending on memory size). The user has the possibility to identify various DNA zones that present either compact or relaxed properties. *ADN-Viewer* was also interfaced with a genomic database containing all living organisms presently sequenced and annotated. The database gives us additional information about genes names, genes locations and other genetic objects within a chromosome. In addition, *ADN-Viewer* is also available within immersive virtual environment, providing stereoscopic visualization on large screens and 3D *user-friendly* interface, using powerful navigation, gesture, speech commands... Such multimodal human-computer interface offers intuitive use and is very suitable when processing and managing huge amounts of data, as it is the case of complete chromosomes.

In this paper, we first give some elementary concepts of genomic data, in particular DNA sequences (section 2). Then, a review of various computer-based representations is proposed in section 3. The section 4 describes our virtual modeling of DNA and its multiple representations. In this section, we also propose some new algorithms dealing with scene management in order to handle and visualize, at different levels, huge chromosomes in real-time. The section 5 presents content-based exploration techniques of annotated DNA, in particular about genes information. In section

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[1] http://www.dnatools.org.

[2] http://www.umass.edu/microbio/rasmol/.

[3] http://www.rcsb.org/pdb.

6, we show that a large immersive environment can be very suitable for complex data analysis in the context of scientific visualization.

2. DNA *IN VIVO*

DNA molecule is composed of four elementary molecules (or nucleotides), represented by four letters: A standing for *Adenine*, C for *Cytosine*, G for *Guanine* and T for *Thymine*. But, the DNA has a double-helix structure and is formed of two complementary strands (Watson strand and Crick one) [13]: when a nucleotide *A* is on one strand, a *T* nucleotide must be on the other strand, and when a *C* nucleotide is on one strand, a G nucleotide must be on the other one. Two complementary nucleotides are linked by hydrogen bonds and thus form a base pair (bp) or a plate. A textual DNA sequence represents only one of the two strands.

3. DNA *IN SILICO*

3.1 3D Engine

The 3D engine of our *ADN-Viewer* software takes as input both textual DNA sequence and 3D conformation model, and it outputs 3D co-ordinates of each nucleotide. The 3D conformation model (established by bio-physicists **Erreur ! Source du renvoi introuvable.**) provides, for each dinucleotide (i.e. each succession of 2 nucleotides in the textual sequence), three angular values and a raise translation. The first plate (base pair of two nucleotides) is placed at the origin of the graphical scene. By scanning the textual sequence in a linear way, *ADN-Viewer* computes the position and orientation of one plate, by applying the Equation 1 on the previous plate position and orientation (Figure 1).

$$M_{xy} = T(-\frac{h}{2}) * R_z(\frac{\Omega}{2}) * Q(\sigma, \delta - 90) * R_z(\frac{\Omega}{2}) * T(-\frac{h}{2})$$

Equation 1. The DNA 3D structure equation, where h = 3.39 angstroms, T is a translation along Z-axis, R is a rotation around X-axis and $Q(\alpha, \beta) = Rz(-\beta)*Rx(-\alpha)*Rz(\beta)$.

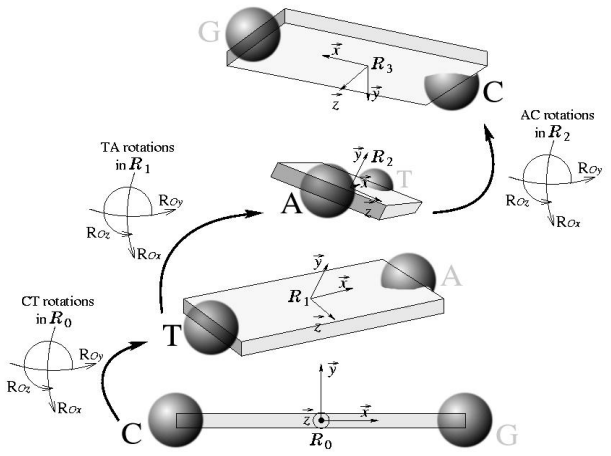


Figure 1. Construction of the DNA 3D trajectory.

3.2 Data Storage

For performance reasons, we have to store the 3D chromosome trajectory resulting from previous computation. Besides, in order to perform quantitative data-processing (compactness, curvature…) on these trajectories, we need to know the spatial position and orientation for each plate. So we store, for each plate, a 3D point (3 double-precision floating points) for position information, and a quaternion [5] (4 double-precision floating points) for the orientation one. By this way, *ADN-Viewer* can process about *n*/50 Mbp, where *n* is the memory size.

4. DNA *IN VIRTUO*

ADN-Viewer offers 3 types of 3D DNA sequence modeling. The sequence may be visualized as a whole, partially (for instance a gene) or at very local scale by displaying all atoms and atomic links of the considered nucleotides.

4.1 Visualization at Chromosome Scale

As we already said, the visualization of a whole chromosome offers us a global point of view about the DNA molecule. The user can identify various areas, where some of them show compactness characteristics and others exhibit more relaxed ones (Figure 2).

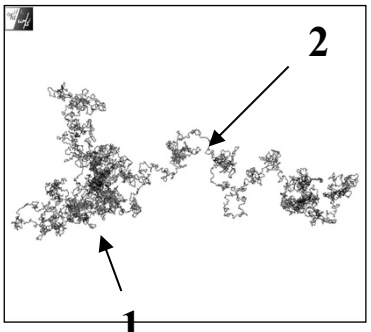


Figure 2. Genomic visualization of *S. cerevisiae chrIII* (~300 Kbp). Area 1 is compact DNA while area 2 is relaxed DNA.

In this genomic modeling, we only display segments linking successive plates. But with a chromosome of 10 Mbp, 10 million segments per frame have to be displayed. In addition, about 15 frames per second (fps) are necessary to have visual fluidity, even if it is well-known that 25 fps rate is better. Consequently, 10.10^6 Mbp * 25 fps = 250.10^6 segments per second have to be displayed. Such a performance is not possible to obtain, even with powerful graphic cards. We will now describe how we tackled this problem by proposing a new filtering algorithm.

4.1.1 Scene management

When a scene contains a lot of objects, it is difficult to have real-time rendering. In order to limit the amount of displayed data, the idea is to display a lot of details of an object when it is closed to the user, and to display a minimum of information when the object is far from him or hidden by another object. To implement these mechanisms, tree structures are often used. In fact, two kinds of scenes exist: hierarchical scenes (for instance urban ones) and non-hierarchical ones (for instance points clouds). For hierarchical scenes, tree leafs are atomic (basical) objects, nodes are subobjects and the root is the complete object. For non-hierarchical scenes that contain several objects, tree leafs represent objects and nodes represent geometric transformations or effects (lights…); concerning non-hierarchical scenes containing only one object (as points cloud) tree leafs and nodes represent squares of scene volume. The tree structure is very useful because each object or scene square can be managed independently to each other and it is very powerful because we can eliminate entire subtrees for

rendering. For non-hierarchical scenes, some algorithms divide the scene into squares to form structure as octree [14], BSP tree or K-D tree and so on. Such algorithms are very efficient but unfortunately very computing expensive.

A scene that contains a 3D DNA sequence is different from the previous ones. It more is assimilated to a points cloud rather than to a urban scene. Nevertheless, on the one hand data are less dense than in an isosurface, the performance gain between the computational time of scene tree and rendering time is strongly reduced. On the other hand, a 3D DNA sequence is a linear strand that constitutes a volume. It means that a 3D point of a DNA sequence is only connected to the previous point and to the next one. So it is connected only to its linear neighbors whereas in an isosurface, a point is connected with its spatial neighbors to define a little triangle or a quad. In order to design a powerful scene management algorithm, the idea was to use the wired structure of a 3D DNA sequence to filter displayed data.

A uniform filtering provides poor rendering performances because of 3D DNA sequence topology. Consequently, we have to apply a non-uniform filtering. The main idea is to detect focused area "on the fly" during the filtering adaptation. For each displayed point, the filtering selects which next point to display by performing a linear function: $d/750$, where d is the distance between the user and the current point, and where 750 has been determined experimentally. In this way, the closer an area is to the user, the more numerous points to display it will contain; and the farer an area is from the user, the less numerous points to display it will contain. At each user movement, the filtering computes the sample step before displaying the chromosome. The gain is difficult to evaluate because it depends if the user is closed to a dense focused area or if he is far from any part of the chromosome. By experience, no visual diseases are observed and all DNA sequences are real-time rendered, even for 20 Mbp molecules.

4.1.2 Optical diseases

Due to the 3D object projection on 2D screen, some undesired optical effects may appear. By applying a fog effect, the depth perception is augmented and the user has a better view of the chromosome volume. However, even with a fog effect, exact perceiving of 3D geometrical feature of a DNA sequence is very difficult, especially due to 2D projection of a 3D object. To overcome such optical effect due to 2D projection, *ADN-Viewer* has a multiview interface to observe the DNA sequence under different angles of view (front perspective view and front, left-side and under orthogonal ones). In section 6, a more interesting solution based on stereoscopic visualization on large screens is described.

4.2 Visualization at Gene Scale

The user has the possibility to select a smaller chromosome part (as gene) in order to observe the double-helix of this part (Figure 3).



Figure 3. Visualization of a gene (~500 bp) with genic modeling. Each nucleotide is represented by a colored sphere.

The non-uniform filtering described in section 2 is applied here again to select which plate will be displayed. In addition, fog effect increases also the 3d structure perception of the gene.

4.3 Visualization at Atomic Scale

Nucleotides are composed of atoms that are not visible with the previous modeling. Here also, the user can select a small part of a gene and observe its nuclear structure (Figure 4).

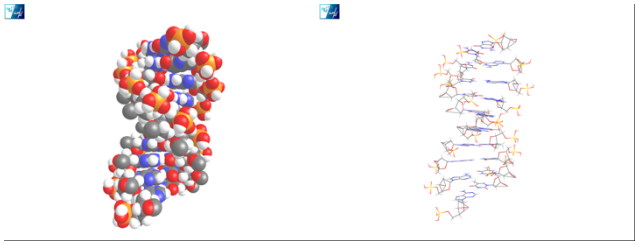


Figure 4. On the left, the full nucleic representation of a gene part (10 bp). On the right, its wired nucleic representation.

Each DNA nucleotide is composed of several linked atoms, and the 3D co-ordinates of each atom are known exactly (X ray experiments). From these biophysical experiments, complete 3D information is obtained and stored in a standard PDB file, containing 3D co-ordinates of all atoms. We have thus the position and orientation for each nucleotide, then geometric transformations are applied and the plate is displayed.

In the full nucleic modeling, each atom is represented by a sphere (about 70 for one plate). There is a lot of sphere intersection and then a lot of hidden faces. Because the positions, centers, radius and atomic links (we use chemical structure of a plate) are all known, we can detect spheres intersections and thus delete all hidden triangles from the data structure. Because only four kinds of plate (A-T, T-A, C-G and GC) exist, the algorithm is applied in the initialisation phase. Hence, we obtain a reduction of about 30 per cent of computed data.

Because our topological database is static, we can do some rendering optimizations. With recent *n*Vidia® graphic cards, we can use OpenGL® **Erreur ! Source du renvoi introuvable.** extensions (**glLockArraysEXT** and **glDrawElements**) that store 3D data directly in video memory and compile structure to optimize the rendering time. With this extension, on a scene that contains 35000 triangles, we obtained a gain of 30 per cent in comparison to previous performances.

5. GENOMIC CONTENT-BASED EXPLORATION

The trajectory is the basic information of DNA sequence. However, a chromosome could be modeled by a collection of genes and intergenic areas. Moreover, several biological signals, patterns or motifs constitute the functional content of these chromosomes. We will then describe how we represent these information by offering the user with genomic content-based exploration.

5.1 Interfacing ADN-Viewer with Genomic Database

We have developed a genomic database (*GenoMEDIA* [3]) that stores annotated DNA sequences: especially information about DNA sequences and genes positions and names. Thanks to SQL queries, *ADN-Viewer* can download any requested annotated sequences in order to augment the visualization (Figure 5). Such augmented representation is very useful for biologists in order to perform supplementary genomic studies. For example, two distant genes in textual DNA sequence could be closed in 3D space, due to the DNA rolling up.



Figure 5. 3D Visualization of *Scerevisiae chrMT* (~50 Kbp). Genes are displayed in white and intergenic areas in black.

The filtering process described in section 4.1.1 does not care of genes positions within the chromosome. In particular, important information about gene start and stop could not be displayed. Genes are displayed with different colors than intergenic areas. So the filtering provides a graduation of colors at gene positions, and this disturbs the user when he wants to observe the exact genes positions in genomic visualization. To overcome this problem, we must force the display of the first and last points of each gene. In addition, with this representation, the user can select any gene by mouse pointing detection mapped in 3D space (Figure 6).

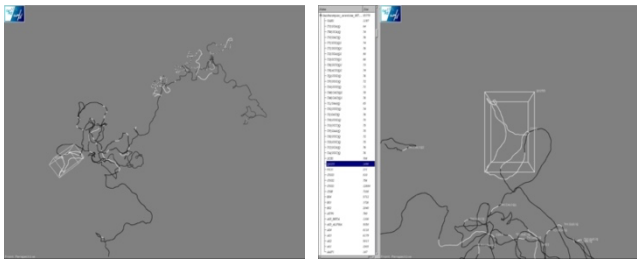


Figure 6. Gene designation by mouse pointing (on the left) and selection by mouse clicking (on the right).

Biologists asked also to have the possibility to display several chromosomes at the same time, which is useful to compare all chromosomes from one eukaryote's genome (17 for *Saccharomyces cerevisiae*), or to compare several chromosomes coming from various organisms. So, *ADN-Viewer* offers the visualization of several chromosomes. Each chromosome has its own model view matrix and thus can be manipulated in an independent manner from others and display additional information. This multi-chromosomes visualization also allows bioinformatics comparison of chromosomes in terms of compactness or other geometrical features. However, this visualization is not exploitable and the genomic analysis becomes unavailable for biologists. Two limitations could be listed: on the one hand, a classical desktop screen has very limited space, and on the other hand, even if the objects are 3D modeled, only 2D projections are perceived on these screens. Stereoscopic perception could help us to overcome such limitations.

6. IMMERSIVE EXPLORATION

Of course, we could have some 3D perception when objects move (manipulations). Nevertheless, the main perceptual tool that provides 3D visualization is a stereoscopic one. Human beings obviously perceive 3D real objects, thanks to binocular vision. So, when the user observes 3D objects on 2D screen, the depth perception is appreciably affected. Besides, if the user interacts with objects with classical devices (keyboard and mouse), he cannot manipulate these 3D objects in an easy way. Classical human-machine interaction paradigms are well adapted for desktop environment. However, they are rapidly useless when 3D objects are manipulated in 3D space. The stereoscopy mechanism and large visualization area make it possible to significantly decrease the gap between the virtual space and the real one **Erreur ! Source du renvoi introuvable.**. For these reasons, *ADN-Viewer* was integrated within the LIMSI Virtual Reality platform (VENISE[4]). This immersive environment platform is equipped with two components: hardware part and software architecture. For the hardware one, we use two rear-projected orthogonal screens (2mx2m size) and a SGI Onyx2® as graphic computer. This hardware device is managed by a middleware developed at LIMSI-CNRS: *EVI3D* **Erreur ! Source du renvoi introuvable.**. This middleware includes two different parts. The first part handles the geometric kernel that manages model views and projections on screens. The second part (*VEServer*) manages various 3D and immersive devices as data glove for gesture recognition, speech recognition system, 3D position and orientation tracking system, immersive mouse… The DNA chromosomes exploration and perception are significantly increased (large screens, stereoscopy, 3D human-machine interfaces, multimodal interaction...) allowing the user to focus completely on his task. Another powerful interaction potential is the interfacing of the vehicle paradigm *HCNav* [1] with *ADN-Viewer*. It consists in navigating in the virtual world with free-hand interaction. Thanks to the tracking of head position and orientation (captured by Flock of Birds® fixed on stereoscopic glasses), the user drives the vehicle in virtual world by his head and body movements. This tracking is also used to adapt dynamically the stereoscopy according to the user's position and orientation. Such navigation offers us two advantages: the user does not need to use the mouse to observe a chromosome any longer, and he can use his hands for other tasks (Figure 7) as gene designation or selection.

[4] http://www.limsi.fr/Recherche/ActionVenise.



Figure 7. *HCNav* system allows free-hand interaction.

For other interaction types (grasping, selection...), many interfacing devices could be used, such as speech for commands or data glove for manipulating objects. *ADN-Viewer* often uses a 3D pointing device: the WANDA™. It has proved very useful to draw a virtual ray having for origin the WANDA™ and for direction the orientation of the device. We can thus easily point or select any object while navigating.

7. CONCLUSION AND FUTURE WORK

In this paper, we addressed the potential offered by Virtual Reality and scientific simulation for 3D modeling and immersive visualization of huge genomic sequences. Advanced work on 3D data modeling and structuring was proposed. The representation of DNA 3D structure allows biologists to observe and analyze genomes in an interactive way at different levels: from gene to chromosome. We developed a powerful software platform that provides is used for sequences analysis: *ADN-Viewer*. This software handles huge masses of data in real-time, we designed various scene management new algorithms and immersive human-computer interaction for *user-friendly* data exploration.

In the future, a first work will deal with the modeling and visualization of whole DNA sequence annotated content (genes, promotors, enhancers, transposons, introns, exons…). The problem is how we can visualize intrinsic 3D information (DNA double-helix) with textual annotations that are not 3D (names, biological origin, function…). In immersive environment, all visualized objects have to be in 3D in order to avoid biologist's cognitive charge if he must alternate between 2D flat text and 3D objects.

A second work will be to investigate the use a PC-based graphical cluster of PCs to display very huge DNA sequences (several hundreds of millions of base pairs, for plants, animals and human organisms). We will be here in front of some databases and applications distribution. We have to design a smart collaboration of the cluster nodes.

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